COMPARING TWO MEANS

DR. HAMMOURI

Introduction

What if we want to compare the mean of some quantitative variable for the individuals in Population 1 and Population 2?

Our parameters of interest are the population means μ_1 and μ_2. The best approach is to take separate random samples from each population and to compare the sample means.

Suppose we want to compare the average effectiveness of two treatments in a completely randomized experiment. We use the mean response in the two groups to make the comparison.

Population or treatment	Parameter	Statistic	Sample size
1	μ_1	$\bar{x}_1$	n_1
2	μ_2	$\bar{x}_2$	n_2

The Sampling Distribution of a Difference Between Two Means

Both $\bar{x}_1$ and $\bar{x}_2$ are random variables. The statistic $\bar{x}_1 - \bar{x}_2$ is the difference of these two random variables. we learned that for any two independent random variables X and Y,

$$\mu_{X-Y} = \mu_X - \mu_Y \quad \text{and} \quad \sigma^2_{X-Y} = \sigma^2_X + \sigma^2_Y$$

The Sampling Distribution of the Difference Between Sample Means

Choose an Simple Random Sample(SRS) of size n_1 from Population 1 with mean μ_1 and standard deviation σ_1 and an independent SRS of size n_2 from Population 2 with mean μ_2 and standard deviation σ_2.

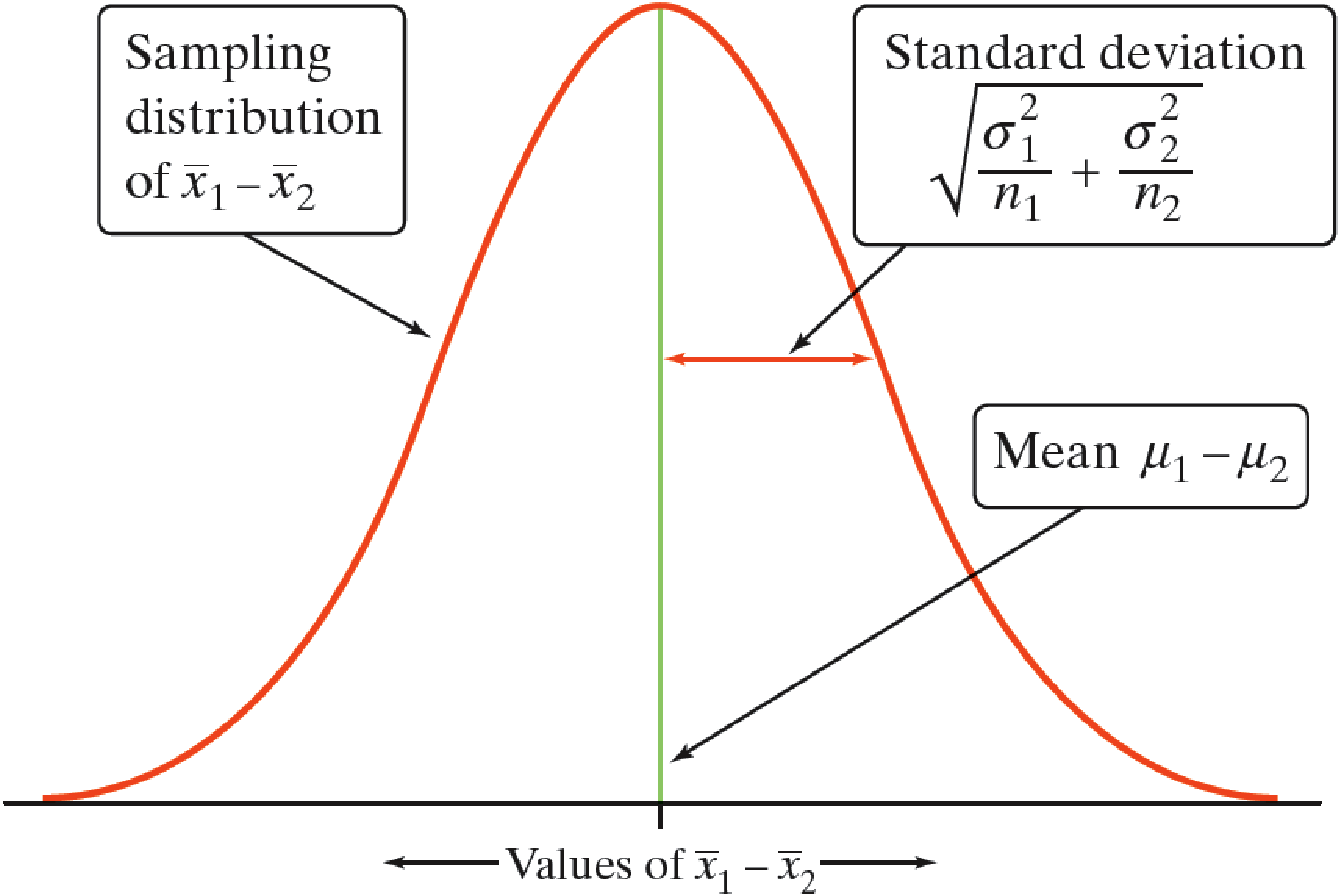
Shape When the population distributions are Normal, the sampling distribution of $\bar{x}_1 - \bar{x}_2$ is approximately Normal. In other cases, the sampling distribution will be approximately Normal if the sample sizes are large enough ($n_1 \geq 30, n_2 \geq 30$).

Center The mean of the sampling distribution is $\mu_1 - \mu_2$.

Spread The standard deviation of the sampling distribution of $\bar{x}_1 - \bar{x}_2$ is

$$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

The Sampling Distribution of a Difference Between Two Means



The Two-Sample *z* Statistic

When data come from two random samples or two groups in a randomized experiment, the statistic $\bar{x}_1 - \bar{x}_2$ is our best guess for the value of $m_1 - m_2$.

the standard deviation of the sampling distribution of $\bar{x}_1 - \bar{x}_2$ is:

$$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^{\ 2}}{n_1} + \frac{\sigma_2^{\ 2}}{n_2}}$$

If the Normal condition is met, we standardize the observed difference to obtain a *z* statistic that tells us how far the observed difference is from its mean in standard deviation units.

Confidence Intervals for $\mu_1 - \mu_2$

Two-Sample *t* Interval for a Difference Between Two Means

When the conditions are met, an approximate % confidence interval for $\mu_1 - \mu_2$ is

$$(\bar{x}_1 - \bar{x}_2) \pm z^* \sqrt{\frac{\sigma_1{}^2}{n_1} + \frac{\sigma_2{}^2}{n_2}}$$

Significance Tests for $\mu_1 - \mu_2$

An observed difference between two sample means can reflect an actual difference in the parameters, or it may just be due to chance variation in random sampling or random assignment. Significance tests help us decide which explanation makes more sense.

The alternative hypothesis says what kind of difference we expect.

$$H_a: \mu_1 - \mu_2 > 0, H_a: \mu_1 - \mu_2 < 0, \text{ or } H_a: \mu_1 - \mu_2 \neq 0$$

The null hypothesis.

$$H_a: \mu_1 - \mu_2 \leq 0, H_a: \mu_1 - \mu_2 \geq 0, \text{ or } H_a: \mu_1 - \mu_2 = 0$$

Significance Tests for $\mu_1 - \mu_2$

To do a test, standardize $\bar{x}_1 - \bar{x}_2$ to get a two-sample z statistic:

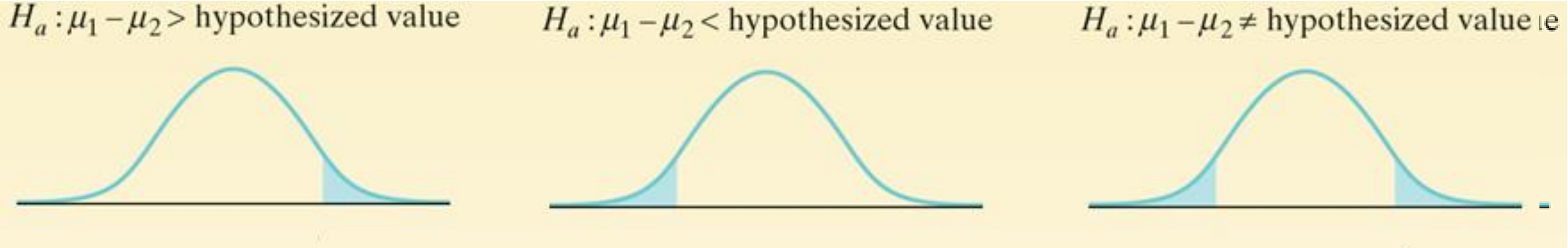
$$\text{test statistic} = \frac{\text{statistic} - \text{parameter}}{\text{standard deviation of statistic}}$$

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^{\ 2}}{n_1} + \frac{\sigma_2^{\ 2}}{n_2}}}$$

use the z table to find the critical value

$z_{\alpha/2}$ or z_α

Then compare as in one sample and reject or accept H_a



Two Sample z-Test for the Means

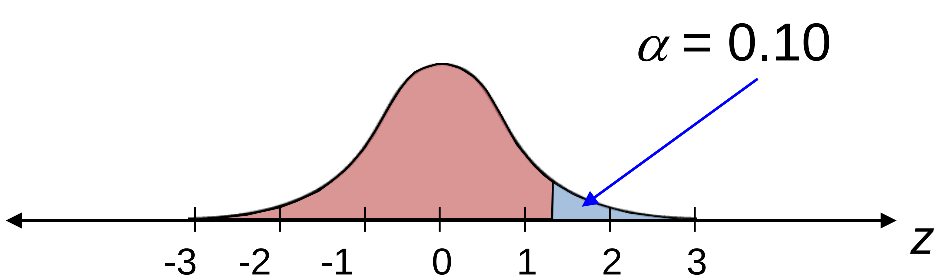
Example:

A high school math teacher claims that students in her class will score higher on the math portion of the ACT then students in a colleague's math class from another state. The mean ACT math score for 49 students in her class is 22.1 and the population standard deviation is 4.8. The mean ACT math score for 44 of the colleague's students is 19.8 and the state population standard deviation is 5.4. At $\alpha = 0.10$, can the teacher's claim be supported?

Then find the 90% CI for the difference?

H_0: $\mu_1 \leq \mu_2$

H_a: $\mu_1 > \mu_2$ (Claim)



$z_0 = 1.28$

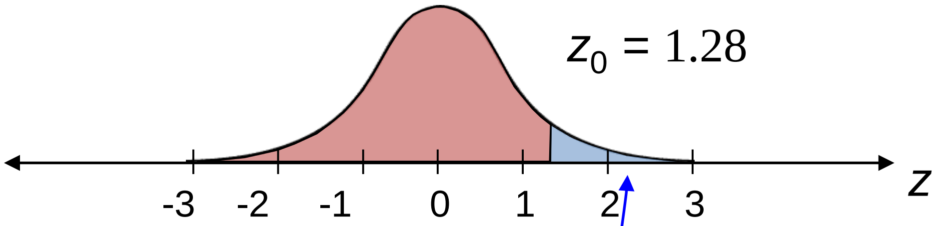
Continued.

Two Sample *z*-Test for the Means

Example continued:

H_0: $\mu_1 \leq \mu_2$

H_a: $\mu_1 > \mu_2$ (Claim)



The standardized error is

$$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} = \sqrt{\frac{4.8^2}{49} + \frac{5.4^2}{44}} \approx 1.0644.$$

The standardized test statistic is

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{x}_1 - \bar{x}_2}} = \frac{(22.1 - 19.8) - 0}{1.0644} \approx 2.161$$

accept H_a.

There is enough evidence at the 10% level to support the teacher's claim that her students score better on the ACT.

Two Sample z-Test for the Means

Example continued:

The standardized error (standard deviation) is

$$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} = \sqrt{\frac{4.8^2}{49} + \frac{5.4^2}{44}} \approx 1.0644.$$

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} * \sigma_{\bar{x}_1 - \bar{x}_2}$$

$$(2.3) \pm 1.64 * 1.0664$$

(0.5511, 4.0489)